

#0 Electron at Metal

- Drude model**
- Basic assumptions**
- (a) Electrical resistance approximation.
 - (b) Electrical resistance results from scattering (all information is lost upon scattering).
 - (c) Relaxation-time approximation.
 - (d) Thermal equilibrium.
- Without electric field** Free electron moving randomly; no net current. Or the drift velocity $v \ll v_{\text{rms}}$.
- EOM**
- (a) Let τ be the scattering time and $1/\tau$ be the scattering rate.
 - (b) In dt , a fraction $\frac{dt}{\tau}$ of the electrons are scattered with zero average velocity.
 - (c) The rest part $1 - dt/\tau$ are accelerated by external force $-eE$.
 - (d) In $t \sim t + dt$, the EOM (omit high-order terms)
- $$m \frac{dv}{dt} = -m\nu/\tau - eE$$
- At steady state ($\frac{dv}{dt} = 0$) The drift velocity
- $$v_d \equiv \langle v(t) \rangle = -e\tau E/m = -\mu E$$
- where mobility $\mu = \frac{e\tau}{m} = \frac{e}{m\nu}$. For a given field, a higher μ means the electrons will move faster. The conductance
- $$\sigma = j/E = -en\tau v_d/E = ne\mu = ne^2 n\tau/m^2.$$
- Here, $\tau = 1/\nu_{\text{rms}}$ is usually independent of E .

Classical Hall effect

- (a) Classically, electrons experience $F = -e(\mathbf{E} + \mathbf{v} \times \mathbf{B})$
 - (b) Cyclotron motion, $F = 0$
- $$-e(\mathbf{E} + \mathbf{v} \times \mathbf{B}) = 0, \quad \omega_c = eB/m^*, \quad r_c = v_{\text{rms}} \omega_c^{-1}/B.$$
- Classically, any orbit size is allowed.

- Drude model with a magnetic field**
- Consider Drude model with applied electric and magnetic fields
- $$m^* \dot{\mathbf{v}} = -e(\mathbf{E} + \mathbf{v} \times \mathbf{B}) - m^* \nu \mathbf{v}$$
- Taking the steady state, with Ohm's rule $j = -nev = \partial E$
- $$\dot{\mathbf{v}} = \begin{pmatrix} \frac{1}{m^*} \frac{\partial E}{\partial v_x} & \frac{1}{m^*} \frac{\partial E}{\partial v_y} & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \dot{\mathbf{v}} = \dot{\mathbf{p}} = \dot{\mathbf{p}}_0 + \dot{\mathbf{p}}_1, \quad \dot{\mathbf{p}}_0 = \frac{e\tau}{m^*} \mathbf{E}$$

- The Hall coefficient $R_H = \rho_{xy}/B = -1/n\mu$.
- Drawbacks**
- (a) Existence of insulators.
 - (b) Temperature dependence of σ .
 - (c) Heat capacity $C_V^D = \frac{3}{2}nk_B$ does not match $C_V \approx 10^{-4}nk_BT$.

- Summertime model for electrons in metals**
- (a) Electrons treated quantum mechanically: Quantize the energy of electron gas.
 - (b) The electronic velocity distribution is given by the quantum Fermi-Dirac distribution.
 - (c) Pauli exclusion: At most one electron can occupy any single electron level.

- Electron wave**
- Schrodinger equation for free electron $\nabla^2 \psi(\mathbf{r}) = -\frac{2mE}{\hbar^2} \psi(\mathbf{r})$. The solution is a plane wave with the energy
- $$\psi(\mathbf{r}) = \frac{1}{\sqrt{V}} e^{i(\mathbf{k} \cdot \mathbf{r} + \omega(\mathbf{k})t + \phi)}$$

- is used. For Simple cube, $R_n = (\pm a, 0, 0)$, $(0, \pm a, 0)$, $(0, 0, \pm a)$, $E(k) = E_0 - \Delta E - 2A[\cos(k_x a) + \cos(k_y a) + \cos(k_z a)]$.
- In the first $2B$ of simple cube, there are several high symmetric points $\Gamma = (0, 0, 0)$, $M = (\pi/a, \pi/a, 0)$, $R = (\pi/a, \pi/a, \pi/a)$, $X = (\pi/a, 0, 0)$. Then $\Delta E_{\text{max}} = 12A$. Near the band bottom
- $$E(k) \approx E_0 - \Delta E - 2A(3 - \omega^2 k^2/2).$$

- The effective mass $m^* = \hbar^2/(2Aa^2)$.
- Comparison of NFE and TBM**

- (a) Nearly free electron model: Good for nearly free electrons (conduction bands).
- (b) TB model for tightly bound electrons (valence bands).

#8 Band Theory II

- Fermi surface**
- (a) Surface in k -space separating occupied and unoccupied states at $T = 0$.
 - (b) Exists only in metals.
 - (c) Shape depends on band structure; volume depends on electron density.

- Effective mass**
- (a) Defined by $m^* = \hbar^2/(E/\partial k^2)$.
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- (c) Can be anisotropic (tensor) and even negative (holes).

- Electron and hole** Missing electron in valence band acts as positive charge carrier with wavevector $\mathbf{k}_h = -\mathbf{k}$, and same velocity as the missing electron.

- Metal, insulator, semiconductor**
- (a) Metal: partially filled band.
 - (b) Insulator: large band gap, filled valence band.
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- Direct vs indirect band gap**
- (a) Direct: valence band maximum and conduction band minimum at same k .
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- Optical properties** Photon absorption possible if $\hbar\nu \geq E_g$.
- (a) Direct gap materials have strong absorption.
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- Excitons** Bound electron-hole pair due to Coulomb attraction. Energy levels below band edge: $E_n = E_g - R_{\infty}/n^2$, where R_{∞} is exciton Rydberg.

- #9 Physics of Semiconductors**
- Intrinsic semiconductors** For Pure semiconductor, carrier concentrations $n = p = n_i$. At finite T ,
- $$n = N_c e^{-(E_c - E_f)/k_B T}, \quad p = N_v e^{-(E_f - E_v)/k_B T},$$
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- Born-Karman periodic boundary conditions**
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